



Submitted To:

**DEPARTMENT OF COMPUTER SCIENCE &
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Certificate

Is a Bonafide work carried out by **MR. MANOJKUMAR PATIL** Reg no. **315CS20012** in partial fulfilment for the course of **COMPUTER SCIENCE & ENGINEERING** during the year 2022-23. It is certified that all corrections/suggestions indicated for internal assessments have been incorporated in the report deposited in the Department library. The project Synopsis has been approved as satisfies the academic requirements in respect of project synopsis report prescribed for the Diploma.

Guide

Program Co-Ordinator

Principal



A Project Synopsis on
"USB Security Key"

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Abstract

To design and implement a complete security platform for USB flash disks due to the popularity of this device in exchanging data, it is a complete system security solution as it concerns the thumb drive due to the manipulation of I/O operation not the file system. USB flash disks have been the major threat for computer system beside the internet threats where viruses can spread from computer to computer or from computer to network.

USB complete security system presented by this paper is composed of three essential elements: kernel filter driver which will be installed in USB device driver stack to intercept all exchanged packets and send it to encryption unit, kernel level encryption/decryption unit and configuration unit.

In contrary to most USB security modules the system presented by this paper will store only the round number of the key generator with the encrypted data. Round number will be coded using MD5 algorithm to increase the immunity of attacking data stored in the flash disks.

Introduction

USB memory sticks can be found almost everywhere. Today, they can be seen as the replacement for floppy-disks, ZIP-drives and all that kind of media. A USB device is indeed a useful, economical way to transfer data. In fact, according to Gartner IT research and advisory company, there were roughly 222 million USB devices shipped in 2009. However, a recent study shows that though USB devices are a convenient means of transferring information, they can also serve as channels to transmit potential threats.

Sunnyvale (California, US) based security firm, Narus Inc. recently listed the 10 "cyber threat trends" for 2011 and beyond, claiming that as attacks through USB becomes inexpensive and data is circulated through them at various public and private locations, the probability of computer Trojans and other malicious software is increasing.

According to Panda's report, 25% of worm based malware spreads through the USB drives. Even more, most of viruses are designed to spread via USB drives. Security Company confirms that cybercriminals are very persistent and put a lot of efforts to make user's life impossible.

25% of latest created malware are configured to enter the system through portable storage devices, usually USB drives. Luis Corrons, the technical director of PandaLabs, the research arm of Panda Security comments: „Much of the malware in circulation has been designed to distribute through these devices. Not only does it copy itself to these gadgets, but it also runs automatically when a USB device is connected to a computer, infecting the system practically transparently to the user.

Diagram

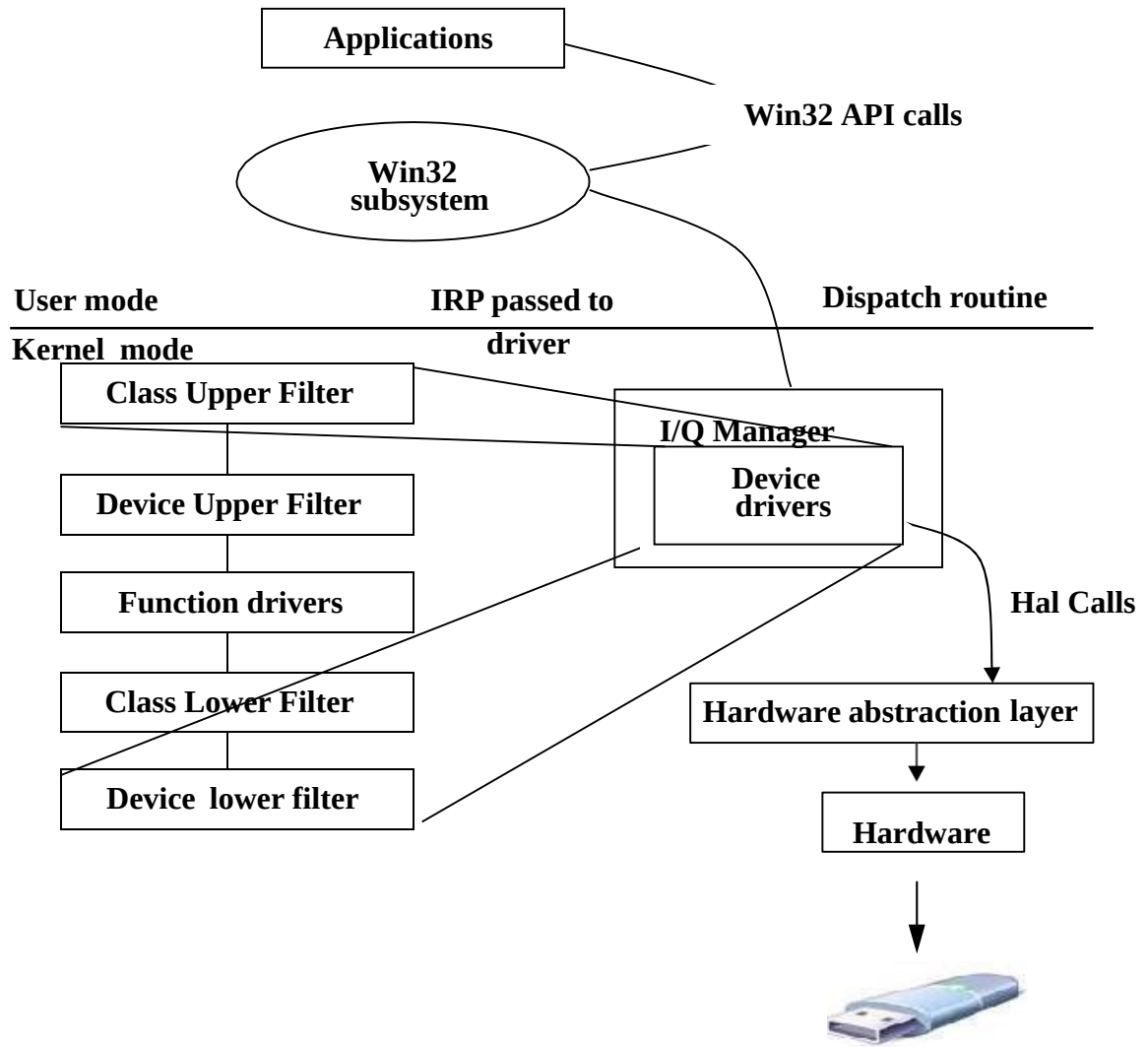
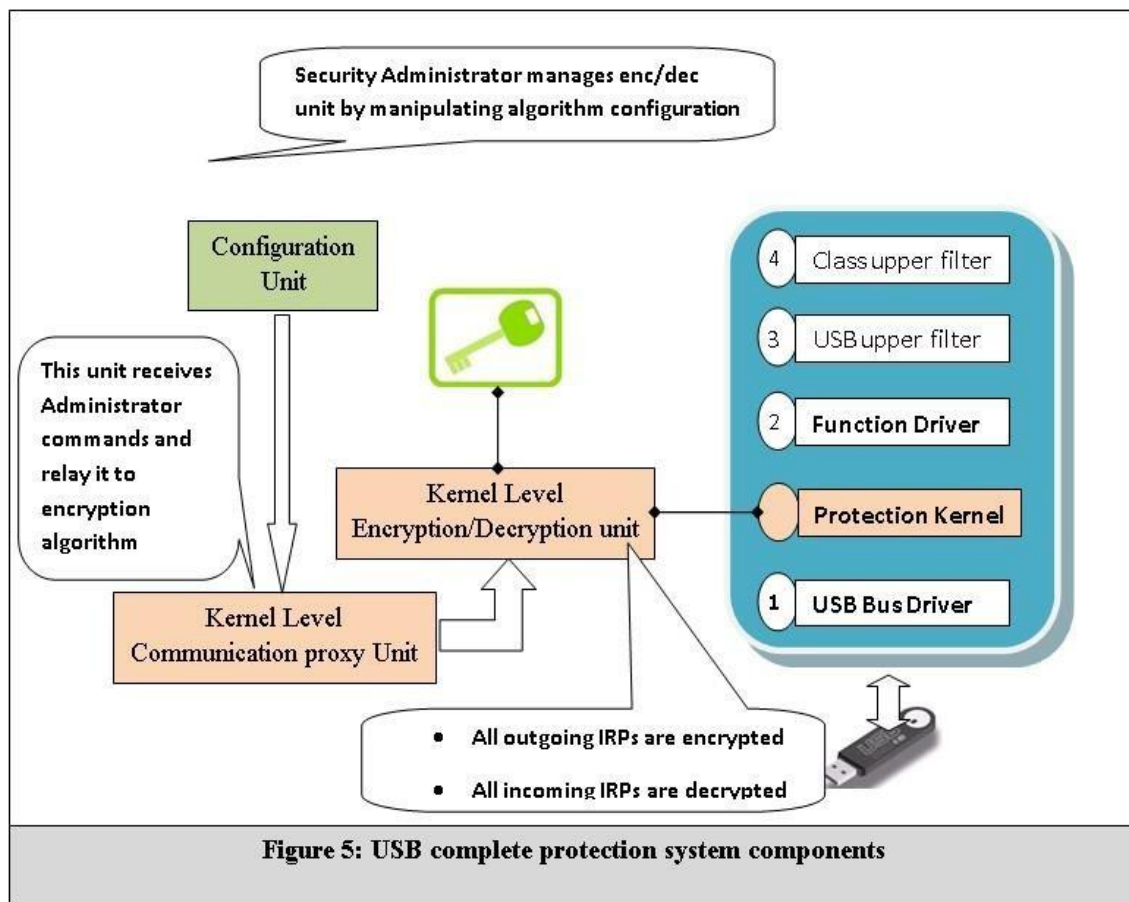


Figure (5) shows the main components of USB complete protection system.



Advantages of USB device management

Securing and managing your USB devices in your corporate network using an enterprise USB device management software can yield many benefits. Some of them are--

- Prevent data theft and data loss that can result due to employees using their personal USB sticks.
- Stop booby-trapped USB devices (devices laced with a worm or virus) from inflicting damage to corporate devices.
- Minimize threats arising from malicious program or code and strengthen corporate security.
- Enterprise USB management software filters out unidentified and unauthorized devices from connecting to your enterprise.

USB Device Audit

IT teams face the cumbersome task of analyzing the usage of pen drives, CDs, or any other portable storage devices that might jeopardize your company's data or security. It is always better to record these activities for future auditing using a USB device manager. Endpoint Central's USB security management software helps track the usage of said USB devices and provides you all the necessary information such as:

- Device name
- User name
- Device type
- Time duration
- Manufacturer and more.

USB Alert

Once a USB device has been restricted, USB alert settings can be configured to notify users whenever they plug in a restricted device. This way, IT teams can leverage Endpoint Central's USB device control features can ensure that the end user complies with the USB restriction policy set up by the organization.

Software Requirements :

- OS Platform : Windows
- Programming Language : Python
- Database : MySQL

Minimum Hardware Specifications :

- Processor : P4 processor or higher versions
- RAM : 512MB or more
- Hard Disk Capacity : 2 GB or more